Synheart Kinematic RulePack

Mathematical documentation of the deterministic kinematic axes

Bemnet Mussa · Synheart AI

July 1, 2026

Contents

Key Terms & Notation **3**

1 Activity State **5**
1.1 Theoretical Basis 5
1.2 Input Signals 6
1.3 Rule Formula 6
1.4 Parameter Notes 6
1.5 Confidence 7
1.6 Interpretation Scale 8
1.7 Context Applicability 8
1.8 Worked Example: Resolving a Jog in the Overlap Band 8
1.9 Implementation Notes 9

2 Movement Regularity **11**
2.1 Theoretical Basis 11
2.2 Input Signals 12
2.3 Rule Formula 13
2.4 Parameter Notes 13
2.5 Confidence 14
2.6 Interpretation Scale 14
2.7 Context Applicability 14
2.8 Worked Example: Steady Commute Walk 15
2.9 Implementation Notes 15

Key Terms & Notation

This front-matter defines the vocabulary shared across every axis chapter, so the chapters themselves stay focused on their own rule. Terms are grouped: signal concepts, then the provenance tags used in the Parameter Notes, then the mathematical notation.

Signal concepts

Term	**Meaning**
Window / step	The short slice of signal analysed at once (e.g. 2.56 or 5 s). The step is how far the window advances each time; "50% overlap" means it advances by half a window, so a new reading appears twice per window length.
Sampling rate F_s	Samples captured per second. Everything here is resampled to 50 Hz.
Magnitude $\lvert a \rvert$	The length of the 3-axis vector, $\sqrt{x^2 + y^2 + z^2}$. Collapsing the axes into one number makes it *orientation-invariant*.
Orientation-invariant	A feature whose value does not change when the device is rotated — so it behaves the same in any pocket or wrist orientation.
Body (gravity-removed) acceleration	Acceleration with the constant 1 g pull of gravity subtracted, leaving only real motion. Taken from a linear-acceleration sensor, or by low-pass-filtering the raw accelerometer.
Movement vs energy	`movement` = the standard deviation of the magnitude (how much it *varies*); `energy` = the mean magnitude (how much acceleration *on average*).
Cadence	Step/stride frequency in Hz, read from the autocorrelation.
Autocorrelation / dominant period	How strongly a signal repeats itself at a given time-lag; the dominant period is the lag of the strongest repeat (the stride, for gait).
Threshold / gate / cut-point	A fixed value the rule compares against to make a yes/no decision.
Confidence proxy	A number in $[0.5, 1]$ giving distance from the decision boundary: 0.5 = right on the boundary (most uncertain), 1 = far from it. It is *not* a calibrated probability.
Tilt	The angle of the device relative to gravity (used by Postural State).
De-rotation	Rotating raw sensor readings into an earth-fixed frame before comparing them (used by Locomotion State).

Provenance tags

Each constant in a Parameter Notes section carries one of these tags, marking honestly where the number comes from.

Tag	Meaning
THEORY	Taken directly from a published method.
DATA	Computed from the data by a formula.
FITTED	Read off the observed feature distributions, then validated on held-out subjects — a human choice informed by the data, not an optimizer's output.
CONV	A standard or physiological choice, not tuned.
PROV	Provisional: data-informed but not locked (e.g. placement-specific). Validate before trusting.
DESIGN	An engineering construction (e.g. the confidence formula).

Notation

Symbol	Meaning
F_s	Sampling rate (50 Hz).
N	Window length in samples (128 for Activity, 250 for the 5 s axes).
a_{body}	Body (gravity-removed) acceleration vector (m/s^2).
$s = \lvert a_{\text{body}} \rvert / g$	Acceleration magnitude in units of g.
$k,\ k^\star$	Autocorrelation lag (samples); $k^\star$ = the dominant-period lag.
$\rho(k)$	Normalized autocorrelation at lag k (in $[-1, 1]$).
n_{cyc}	Number of motion cycles in the window, $N/k^\star$.
$T_\bullet$	A named threshold (e.g. T_{STILL}, T_{RUN}, T_{GYRO}).
$\text{clamp}(x, 0, 1)$	Clip x to the range $[0, 1]$.
conf	Confidence, in $[0.5, 1]$.

Chapter 1

Activity State

Activity State estimates *the type and intensity of whole-body movement* — sedentary, walking, running, or cycling — from the strength and frequency of acceleration. It is theoretically distinct from **Postural State** (which measures body orientation, not whether or how the body is moving) and from **Movement Regularity** (which measures how rhythmic a motion is, not what kind it is or how energetic). This is the foundational axis: it labels *what the body is doing*, and the other axes either build on that label or resolve the questions it deliberately leaves open.

Scope. This axis reports the dominant activity *class* over a short window, not a continuous intensity. Standing folds into *sedentary* on purpose — posture is a separate question, delegated to Postural State. Cycling, by contrast, is kept as its own class because the body is working hard while producing little linear acceleration: without a dedicated (gyroscope) cue it would read as sedentary, and its raised heart rate would then be misattributed to stress rather than exercise. The confound most likely to occur is **walking versus running**: their energy distributions genuinely overlap, and the split in that overlap rests on cadence and on placement-specific thresholds. Cycling can only be detected when a gyroscope is present; without one it reads as sedentary.

1.1 Theoretical Basis

The rule is an *accelerometer intensity cut-point* classifier, the decades-old method of the physical-activity-monitoring literature, extended with a cadence tie-breaker and a rotational cue for cycling. The core feature — the standard deviation of the acceleration magnitude — is the same construct as **MAD** (Mean Amplitude Deviation, Vähä-Ypÿä et al., 2015) and **ENMO** (Euclidean Norm Minus One, Hildebrand et al., 2014, the GGIR default), and the cut-point idea traces to activity-count thresholds (Freedson et al., 1998; Troiano et al., 2008). What is *directly evidenced* by that work is the use of a magnitude-intensity statistic against fixed thresholds. The *engineering adaptations*, flagged below, are the specific threshold values (re-derived on MotionSense), the cadence tie-breaker inside the overlap band, and the gyroscope cycling branch.

- **Acceleration magnitude intensity** — $\text{std}(|a_{\text{body}}|)$ separates still from moving and low- from high-energy gait; it is orientation-invariant, so it does not depend on how the device is carried (Yurtman & Barshan, 2017).

- **Cadence (step frequency)** — where energy alone cannot separate a fast walk from a slow run, step rate can; walking and running occupy different cadence ranges in the gait literature.

- **Rotational energy for cycling** — pedalling produces little linear acceleration but strong limb/segment rotation, so mean gyroscope magnitude detects cycling that the accelerometer misses.

1.2 Input Signals

Signal	Symbol	Description
Body acceleration	a_{body}	Gravity-removed 3-axis acceleration (m/s^2)
Motion size	`move`	$\text{std}(\lvert a_{\text{body}}\rvert/g)$ over the window (g)
Energy	`energy`	$\overline{\lvert a_{\text{body}}\rvert/g}$, mean magnitude (g)
Cadence	`cad`	Step frequency (Hz) from the autocorrelation peak
Gyroscope energy	`gyro`	$\overline{\lvert\omega\rvert}$, mean angular-rate magnitude (rad/s); cycling cue

Window 128 samples (2.56 s at $F_s = 50\,\text{Hz}$), 50% overlap — the shortest of the axes, so activity reacts quickly. No personal baseline: the thresholds are population-derived and frozen.

1.3 Rule Formula

RULE-AS *Activity State Classifier*

Step 1 — Features ($s = |a_{\text{body}}|/g$; cadence over lag band $[10, 64]$):

$$\texttt{move} = \text{std}(s), \quad \texttt{energy} = \overline{s}, \quad \texttt{cad} = \frac{F_s}{k^\star}, \quad \texttt{gyro} = \overline{|\omega|}.$$

Step 2 — Still gate:

$$\texttt{move} < 0.10 \text{ g} \;\Rightarrow\; \textbf{sedentary}.$$

Step 3 — Walk / run split (else, i.e. moving):

$$\begin{aligned} 0.77 \le \texttt{energy} \le 1.07 :&\quad \textbf{running if } \texttt{cad} \ge 1.20 \text{ Hz, else } \textbf{walking} \\ \text{otherwise} :&\quad \textbf{running if } \texttt{energy} \ge 1.00 \text{ g, else } \textbf{walking} \end{aligned}$$

Step 4 — Cycling override (only in the still region, needs gyro):

$$\big(\texttt{move} < 0.10 \;\wedge\; \texttt{gyro} > 0.18 \text{ rad/s}\big) \;\Rightarrow\; \textbf{cycling}.$$

$\boxed{\texttt{label} \in \{\text{sedentary, walking, running, cycling}\}}$

Example (real window, § 1.8): `move` $= 0.54$, `energy` $= 0.91$ (in the overlap band ⇒ energy cannot decide), `cad` $= 1.43 \ge 1.20$ ⇒ **running**.

1.4 Parameter Notes

Still threshold $T_{\textbf{STILL}} = 0.10\,\textbf{g}$ [FITTED, MOTIONSENSE]**.** Below this, magnitude variation is sensor noise, not body motion. The UCI-HAR value (0.05) did *not* transfer because of the unit/placement difference, which is why every threshold here is stated in g on the same body-acceleration scale.

Run energy line $T_{\mathbf{RUN}} = 1.00\,\mathbf{g}$ and overlap band $[0.77,\ 1.07]$ [FITTED / DATA, MOTIONSENSE]**.** Outside the band, mean magnitude alone separates walking from running. Inside the band the two classes' energy distributions touch, so energy is *not* trusted there — the cadence tie-breaker takes over.

Cadence split $CAD_{\mathbf{RUN}} = 1.20\,\mathbf{Hz}$ [PROVISIONAL, MOTIONSENSE]**.** Used *only* inside the overlap band. It is placement-specific: on waist data (HHAR) the autocorrelation fundamental flips between step and stride, so this value does not transfer unchanged. In practice cadence is a *narrow refinement, not a pillar*: it fires only in the overlap band, lifted held-out accuracy by about 1.8 points (95.3 → 97.1%), and is the most fragile part of the rule — an item to re-derive on own-phone pocket data.

Gyro cycling threshold $T_{\mathbf{GYRO}} = 0.18\,\mathbf{rad/s}$ [PROVISIONAL, HHAR]**.** Derived on HHAR (waist); pocket cycling is not yet verified. The branch fires only in the still region so it cannot corrupt walking/running — but on held-out data this threshold both *misses* hard pedalling and *false-fires* on phone-handling while standing (quantified in § 1.5), so it is an open validation item.

Fusion method. This is a *priority decision cascade*, not a weighted sum. Energy decides first; cadence is a *tie-breaker* used only where energy is ambiguous; the gyroscope is an *override gate* confined to the still region. The ordering encodes which cue is trusted in which regime, keeping each class's decision traceable to one deciding quantity.

Tags (FITTED, DATA, PROV, CONV, DESIGN) classify where each number comes from; their meanings are defined in the Key Terms front-matter. Window (128/64) and cadence lag band ($[10, 64]$) are CONV (standard/physiological choices); the confidence formula below is DESIGN.

1.5 Confidence

Confidence is a deterministic *margin proxy* in $[0.5, 1.0]$: how far the deciding quantity sits from the threshold that chose the label.

Step C1 — Deciding margin m (by the branch that fired):

$$m = \begin{cases} (\mathtt{gyro} - 0.18)/0.18 & \text{cycling} \\ (0.10 - \mathtt{move})/0.10 & \text{sedentary} \\ |\mathtt{cad} - 1.20|/1.20 & \text{walk/run inside the band} \\ (\mathtt{energy} - 1.00)/1.00 & \text{running (outside band)} \\ (1.00 - \mathtt{energy})/1.00 & \text{walking (outside band)} \end{cases}$$

Step C2 — Confidence: $\boxed{\text{conf} = 0.5 + 0.5\ \text{clamp}(m,\ 0,\ 1)}$

Confidence ceiling / caveat. This is a *proxy*, not the trained, calibrated confidence. On the source bench (MotionSense) the calibrated model reaches ECE = 0.024; the proxy reads 0.5 at a decision boundary (maximum uncertainty) and 1.0 far from it. Off-source it is less calibrated, so it should be read as a relative distance-to-boundary, not a probability.

Measured per class (median confidence, held-out data): sedentary 0.97, running 0.73, walking 0.66, cycling 1.00 when it fires. Walking is structurally the least confident — it is squeezed between the still line below and the run line above, so it always sits near a boundary. Cycling is confident *when* the gyro cue trips, but that confidence measures distance-past-threshold, not correctness: on held-out data the branch *missed* ∼31% of real cycling (hard pedalling crosses the still gate and reads as walking) and *false-fired* on ∼16% of standing (phone handling pushes gyro past the provisional T_{GYRO}). Both are open items to validate on own-phone pocket data.

1.6 Interpretation Scale

Activity State is categorical; the “scale” is the class set and each class’s signature.

Class	Signature	Meaning
Sedentary	`move` $< 0.10\,\mathrm{g}$	Still or near-still: sitting, standing, resting (posture delegated)
Walking	Moving, below the run line / in-band cadence $< 1.20\,\mathrm{Hz}$	Steady low-to-moderate gait
Running	`energy` $\geq 1.00\,\mathrm{g}$, or in-band cadence $\geq 1.20\,\mathrm{Hz}$	High-energy, fast-cadence gait
Cycling	Still-region body motion with `gyro` $> 0.18\,\mathrm{rad/s}$	Low linear motion but rotational (pedalling); needs a gyroscope

1.7 Context Applicability

Target context: a phone or wearable carried on the body (pocket-scale placement), sampling accelerometer and gyroscope at 50 Hz.

Context	accel	gyro	Reliable?
Phone in pocket (target)	Yes	Yes	Yes — all four classes
Waist / no gyroscope	Yes	No	Partial — cycling reads as sedentary
Wrist / other placement	Yes	Partial	Partial — cadence/gyro thresholds shift

Scientific boundary. Even with ideal data the axis cannot separate *standing* from *sitting* or from sedentary — that needs orientation, delegated to Postural State. The walking/running boundary is a real energy overlap resolved heuristically by cadence, and both the cadence and gyro thresholds are placement-specific rather than universal. Cycling additionally requires a placement that is rotationally active but linearly quiet during pedalling: the torso qualifies (chest), the limbs do not (wrist/ankle keep moving and read as walking), so cycling detection is torso-placement-specific.

1.8 Worked Example: Resolving a Jog in the Overlap Band

Scenario (real captured window). A user jogs at a steady pace, phone in the front pocket. The window below is a genuine capture, not an invented one — MotionSense subject 19 (held out from all threshold derivation), jog trial, 58.9 s in (`jog_9/sub_19.csv`, samples 2944–3071) — run through the live axis. Its energy lands inside the walk/run overlap band, so energy alone cannot decide the class.

Signal	**Value**
`move` (std of s)	0.54 g
`energy` (mean of s)	0.91 g (in band $[0.77, 1.07]$)
`cad`	1.43 Hz
`gyro`	1.87 rad/s

Step-by-step:

$$\texttt{move} = 0.54 \geq 0.10 \Rightarrow \text{not sedentary}$$
$$\texttt{energy} = 0.91 \in [0.77, 1.07] \Rightarrow \text{use cadence, not energy}$$
$$\texttt{cad} = 1.43 \geq 1.20 \Rightarrow \textbf{running}$$
$$\texttt{move} < 0.10 \text{ false} \Rightarrow \text{cycling override does not fire}$$
$$m = |1.43 - 1.20|/1.20 = 0.19, \quad \text{conf} = 0.5 + 0.5 \cdot 0.19 = \mathbf{0.60}$$

Interpretation: the window is classified *running*. Because the energy fell in the overlap band, the decisive driver was the *cadence* (1.43 Hz), not the energy. Confidence is moderate (0.60): the step rate is above but not far above the 1.20 Hz walk/run line, which is exactly the region where this axis is least certain. The gyroscope reads high (1.87 rad/s) but is ignored — the cycling override only fires in the still region, and here the body is clearly moving.

1.9 Implementation Notes

- **Window.** 128 samples (2.56 s) at 50 Hz, 50% overlap — the shortest window in the pack, so the activity label updates fastest. There is an inherent latency: a 2.56 s window means an activity change is only caught within about 1.3–2.6 s (a new reading every ∼1.3 s). This matches the physiology it contextualises (heart rate and affect shift over tens of seconds) but is *not* intended for instantaneous gesture or fall detection.

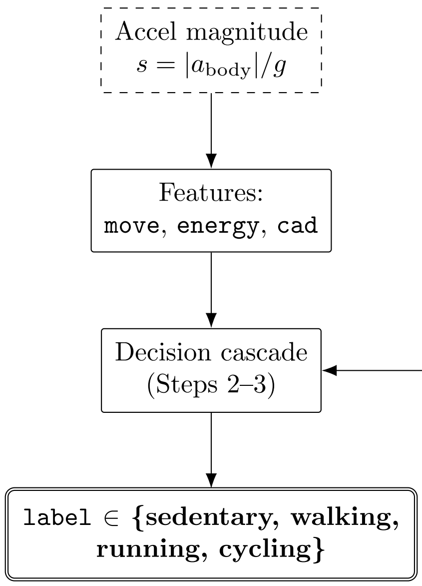
- **Fallback behavior.** No accel/linear stream → null. No gyroscope → the cycling branch is disabled and the reading note says so; the accel classes are unaffected. Cadence still computes from the accelerometer.

- **Relationship to other chapters.** The standing/sitting ambiguity it folds into sedentary is resolved by **Postural State**; its walking/running output gates **Movement Regularity** (only meaningful while moving) and feeds **Locomotion** and the **Overall State** combiner.

- **Validation status.** Headline **97.1%** on MotionSense (phone in pocket, held-out subjects; macro-F1 0.948, well-calibrated ECE = 0.024). The rule then holds *frozen* (nothing re-derived) across two further datasets and sensor types (table below) — notably the walk/run boundary, which HHAR cannot test (it has no running), is cross-validated on PAMAP2. Cycling is the one placement-dependent part. CAD_{RUN} and T_{GYRO} stay provisional; pocket cycling and a phone-in-pocket walk/run set (e.g. WISDM) are the remaining checks.

Dataset	**Sensor / placement**	**Classes**	**Result (frozen)**
MotionSense	phone, pocket	sed / walk / run	97.1% held-out (macro-F1 0.948)
HHAR	phones + watches, waist/hand	sed / walk (no running)	macro-F1 0.996 shared
PAMAP2	research IMUs, wrist/chest/ankle	walk / run	0.89–0.96 acc., all 3 placements
PAMAP2	” (cycling)	cycling	chest recall 0.67; wrist/ankle ∼0.03

PAMAP2 figures use offline per-segment gravity removal; the live on-device per-window approximation lowers only ankle-walking (0.72 vs 0.93) — running and the torso/wrist placements are unaffected.



Formula flow — Activity State: a priority decision cascade over orientation-invariant accelerometer features, with a gyroscope override for cycling.

Chapter 2

Movement Regularity

Movement Regularity estimates *how rhythmic and self-repeating the body's motion is over a short window* — a smoothness score, not an intensity or a posture. It is theoretically distinct from **Activity State** (which measures *how much* motion and of *what type*) and from **Postural State** (which measures orientation relative to gravity). A person can be highly active yet irregular (stumbling, fidgeting) or barely active yet perfectly regular (a slow steady walk); regularity is the axis that separates these, so it is a genuinely non-redundant construct rather than a re-scaling of activity intensity.

Scope. This score reports *whether* motion repeats cleanly, not *why* it does or does not. A low score cannot by itself distinguish a mechanical cause (rough terrain, an obstacle) from a physiological one (tremor, agitation, fatigue). The score is only defined while the body is actually moving rhythmically; when the person is still, or when too few motion cycles fall inside the window, the axis returns **null** rather than a misleadingly "perfect" value. Resolving the cause of irregularity requires the Activity State context and the downstream heart-rate / affective layer.

2.1 Theoretical Basis

The score is the normalized *unbiased autocorrelation* of the acceleration-magnitude signal at its dominant period. This is the trunk-accelerometry gait-regularity index of **Moe-Nilssen & Helbostad (2004)**: for a walking signal, the value of the normalized autocorrelation at the stride lag measures how closely one stride reproduces the next, and the authors note it is cheap enough for portable devices. The parts of the formula *directly evidenced* by that work are the unbiased-autocorrelation estimator and the use of its dominant-period peak height as the regularity index. The remaining choices are *engineering adaptations*, flagged as such below: the clamp to $[0, 1]$, the movement and cycle-count gates, the pocket-placement lag band, and the two-factor confidence.

- **Acceleration magnitude** $|a_{\mathbf{body}}|$ — gravity-removed body acceleration reduced to its Euclidean norm, making the feature orientation-invariant (the same rhythm reads identically regardless of how the phone is rotated in the pocket).
- **Unbiased autocorrelation** — the $1/(N-k)$ normalization corrects the end-of-window sample deficit so that longer lags are not artificially damped (Moe-Nilssen & Helbostad, 2004).
- **Smoothness alternatives (not used, cited for completeness)** — spectral arc length SPARC (Balasubramanian et al., 2012) and log dimensionless jerk (LDLJ) are established

smoothness measures; autocorrelation is chosen here because it is period-aware and matches the gait literature.

- **Affective relevance** — reduced or irregular ambulatory movement is associated with agitation and anxiety in actigraphy studies, which motivates exposing regularity to the downstream affective read.

2.2 Input Signals

Signal	Symbol	Description
Body acceleration	a_{body}	Gravity-removed 3-axis acceleration (m/s^2); linear-accel sensor, or raw accel low-passed at 0.3 Hz
Acceleration magnitude	s	$\lvert a_{\text{body}} \rvert / g$, orientation-invariant motion size (g)
Dominant lag	$k^\star$	Period (samples) of the strongest self-repeat in the window
Activity state	`ACT`	Context gate: regularity is only meaningful while walking/running

Window $N = 250$ samples (5 s at $F_s = 50\,\text{Hz}$), 50% overlap. No personal baseline is required: the score is self-referential within the window.

2.3 Rule Formula

RULE-MR *Movement Regularity Score*

Step 1 — Magnitude and detrend:

$$s(t) = \frac{|a_{\text{body}}(t)|}{g}, \qquad x(t) = s(t) - \bar{s}, \qquad \text{movement} = \text{std}(s).$$

Step 2 — Unbiased, normalized autocorrelation (lags $k = 0 \dots K$, $K = 100$):

$$r(k) = \frac{1}{N-k} \sum_{t=0}^{N-1-k} x(t)\, x(t+k), \qquad \rho(k) = \frac{r(k)}{r(0)}.$$

Step 3 — Dominant period (argmax over the stride band, not the first peak):

$$k^\star = \underset{k \in [10,100]}{\arg\max}\, \rho(k), \qquad n_{\text{cyc}} = \frac{N}{k^\star}.$$

Step 4 — Score:

$$R = \text{clamp}\big(\rho(k^\star),\, 0,\, 1\big).$$

Step 5 — Validity gate (return **null** instead of a score if):

$$\text{movement} < 0.051 \text{ g} \quad (\text{low\_movement}) \quad \text{or} \quad n_{\text{cyc}} < 3.5 \quad (\text{insufficient\_cycles}).$$

$\boxed{R \in [0,1] \;\text{ if valid, else null}}$

Example: steady pocket walk, $\text{std}(s) = 0.37\,\text{g}$, $k^\star = 55$ ($\approx 0.9\,\text{Hz}$ stride), $\rho(k^\star) = 0.71$, $n_{\text{cyc}} = 250/55 = 4.5$. Both gates pass $\Rightarrow R = 0.71$ ("rhythmic").

2.4 Parameter Notes

Regularity estimator [THEORY]**.** The score itself — the normalized unbiased autocorrelation at the dominant period — is taken directly from the Moe-Nilssen & Helbostad (2004) trunk-gait index. It is the one number here that comes straight from published theory; everything else is a gate or a confidence around it.

Lag band $[10, 100]$ **samples** [CONV]**.** At 50 Hz this covers stride periods 0.2–2.0 s (0.5–5 Hz), the plausible human cadence range. Taking the *argmax* over the whole band (rather than the first prominence peak) is deliberate: on pocket data the autocorrelation has strong sub-stride harmonics, and the first peak grabs a half-stride and inverts the rank ordering. This is the FIX 2 correction re-derived during the chest→pocket convergence test.

Movement floor 0.051 **g** [DESIGN]**.** Below this, magnitude variation is sensor noise, not motion, and the autocorrelation of noise produces a spurious high peak. The value is the chest-derived $0.5\,\text{m/s}^2$ floor re-expressed in g (0.5/9.81); expressing it in stream units was FIX 1 of the convergence test.

Minimum cycles 3.5 **strides** [CONV]**.** Fewer than ∼3–4 repeats in the window makes a "regularity" estimate statistically meaningless. 3.5 stride-periods is the gait-literature adequacy minimum, re-expressed for the argmax-selects-stride convention on pocket placement.

Prominence saturation $P_{\textbf{full}} = 1.5$ [DESIGN]**.** The peak prominence at which the confidence prominence term reaches 1; a chosen saturation point, not a measured quantity.

Fusion method. There is *no* multi-signal fusion in the score itself — it is a single autocorrelation read. Fusion appears only in the *confidence* (§ Confidence), which combines two *independent* adequacy factors (enough cycles, and a prominent peak) as a geometric mean, so that either factor being weak pulls confidence down. The score's *separation* (not an absolute accuracy) was validated on PAMAP2/MotionSense; the gates and confidence are DESIGN choices tuned to keep noise from scoring as rhythm. Tag meanings are defined in the Key Terms front-matter.

2.5 Confidence

Step C1 — Cycle adequacy (how far past the minimum the window sits):

$$c_{\text{cyc}} = \text{clamp}\left(\frac{n_{\text{cyc}} - 3.5}{5.0 - 3.5},\, 0,\, 1\right).$$

Step C2 — Peak prominence (p = prominence of the peak at $k^\star$; 1.5 = saturation):

$$c_{\text{prom}} = \text{clamp}\left(\frac{p}{1.5},\, 0,\, 1\right).$$

Step C3 — Final confidence (geometric mean of the two): $\boxed{\text{conf} = \sqrt{c_{\text{cyc}} \cdot c_{\text{prom}}}}$

Confidence ceiling / caveat. This axis reports no labelled accuracy by design — it is validated by *separation* (rank-ordering rhythmic above irregular above still), not against a ground-truth "regularity" label, which does not exist in the datasets. Confidence therefore expresses *internal signal quality* (enough clean cycles), not calibrated correctness, and should not be read as a probability.

2.6 Interpretation Scale

Score	Level	Meaning
0.0–0.2	Erratic	Motion present but not self-repeating (stumbling, fidgeting)
0.2–0.4	Low regularity	Weak, inconsistent rhythm
0.4–0.6	Moderate	Discernible but imperfect rhythm
0.6–0.8	Rhythmic	Clear, steady periodic motion (typical walking)
0.8–1.0	Highly regular	Metronomic, strongly self-repeating (steady running/pace)
null	Not applicable	Still, or too few cycles — no rhythm to score

2.7 Context Applicability

Target context: on-body (pocket / trunk) accelerometry during *sustained locomotion.* The score is most reliable when the Activity State is walking or running.

Context	$\lvert a_{\textbf{body}}\rvert$	cycles ≥ 3.5	Reliable?
Walking / running (pocket or trunk)	Yes	Yes	Yes — full
Cycling / non-gait rhythmic motion	Yes	Partial	Partial (period may fall outside band)
Standing / sitting / lying	No	No	No — returns null

Scientific boundary. Even with ideal data, the score cannot attribute *cause*: it measures the presence of rhythm, not its origin, so it approximates "movement smoothness" rather than directly measuring any affective or neuromotor state. On PAMAP2 (chest) the score separates rhythmic from irregular-but-moving activity, so it is not merely a movement detector; on the pocket, only the still-versus-rhythmic contrast is confirmed so far, and rhythmic-versus-irregular-moving on the pocket is owed to own-phone data.

2.8 Worked Example: Steady Commute Walk

Scenario. A user walks to a bus stop at a relaxed, even pace, phone in the front trouser pocket. Over one 5 s window the pipeline extracts:

Signal	**Value**
std(s) (movement)	0.37 g
Dominant lag $k^\star$	55 samples ($\approx$ 0.9 Hz stride)
$\rho(k^\star)$	0.71
Peak prominence p	0.60

Step-by-step:

$$\begin{aligned}
n_{\text{cyc}} &= 250/55 = 4.5 \ (\geq 3.5,\ \text{movement } 0.37 > 0.051 \Rightarrow \text{valid}) \\
R &= \text{clamp}(0.71, 0, 1) = \mathbf{0.71} \\
c_{\text{cyc}} &= \text{clamp}\big((4.5 - 3.5)/1.5, 0, 1\big) = 0.67 \\
c_{\text{prom}} &= \text{clamp}(0.60/1.5, 0, 1) = 0.40 \\
\text{conf} &= \sqrt{0.67 \cdot 0.40} = \mathbf{0.52}
\end{aligned}$$

Interpretation: $R = 0.71$ falls in the *Rhythmic* band — a clean, steady walking cadence. The dominant driver is the high autocorrelation peak ($\rho = 0.71$); confidence is moderate (0.52), held down by the modest peak prominence rather than by too few cycles.

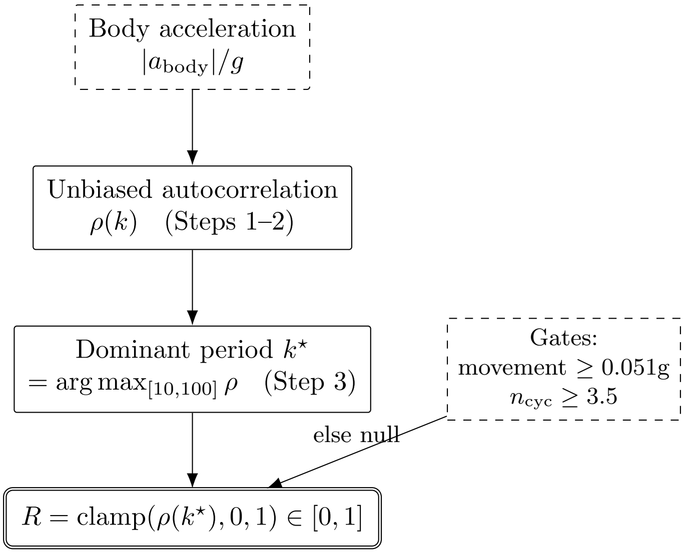
2.9 Implementation Notes

- **Window.** 250 samples (5 s) at 50 Hz, 50% overlap; magnitude is read from the same rolling buffer the other 5 s axes use.
- **Fallback behavior.** No accel/linear stream → null. A window that is > 50% bad samples (NaN or > 16 g), or whose first sample is bad, → null (`window_unreliable`); shorter gaps are forward-filled. If `scipy` is unavailable the score still computes; only the prominence confidence factor degrades.
- **Relationship to other chapters.** Gated by **Activity State** — a regularity number is only surfaced while `ACT` indicates locomotion; on its own, regularity cannot tell moving-irregular from still, which is why the two axes are read together.
- **Validation status.** No labelled accuracy by design (there is no ground-truth regularity label), so validation is by *separation.* On PAMAP2 chest, rhythmic vs irregular ROC-AUC 0.99 (per-subject median 0.99, all 8 evaluable subjects $\geq$ 0.97). It transfers *frozen* to two further datasets and sensor types (table below). **And it is regularity, not movement:** restricted to *moving* windows — so "moving vs still" cannot explain the split — rhythmic vs irregular-but-moving (vacuuming, ironing) still separates at ROC-AUC 0.99: vacuuming

moves yet scores ∼0.25 while walking scores ∼0.94 (regularity correlates with movement only ∼+0.5). This movement-control exists only on PAMAP2; the pocket version is owed to own-phone data.

Dataset	**Placement / sensor**	**Contrast**	**ROC-AUC**
PAMAP2	chest, research IMU	rhythmic vs irregular	0.99
MotionSense	pocket, phone	rhythmic vs still	0.96
HHAR	waist/hand, phone + watch	walking vs still	0.98

MotionSense/HHAR have no irregular-but-moving activity, so cross-dataset confirms rhythmic > still only. HHAR must be resampled *within gap-free spans*: its heterogeneous sampling, interpolated across gaps, otherwise fakes periodicity.



Formula flow — Movement Regularity: one orientation-invariant input, period-aware autocorrelation, a validity gate, and a $[0, 1]$ score.